

Problem Statement:

Modern financial platforms struggle to balance fraud mitigation with user privacy. Online payment scams and credential-harvesting banking frauds are rapidly increasing day by day. Traditional fraud detection engines rely heavily on decrypting and parsing sensitive data like primary account numbers, identifiers, or personal data. This creates significant vulnerability risks during the pipeline transit layer.

Core Challenges:

- ❑ **Escalating Scams:** Growth in sophisticated phishing, social engineering, and rapid account-draining operations.
- ❑ **Privacy Violations:** Ingesting raw PII data risks compliance failures under strict global financial regulations.
- ❑ **High-Fidelity Attacks:** Threat vectors like unverified hardware links or rapid velocity spikes bypass legacy checks.
- ❑ **System Friction:** Rigid legacy fraud systems trigger excessive false positives, creating high customer friction.

Objectives:

- Develop a **Zero-PII Ingestion Pipeline** that tokenizes and strips sensitive consumer credentials at the gateway layer.
- Implement an orchestration engine using **watsonx foundation models** to analyze unidentifiable structural metadata in real time.
- Deploy a deterministic threat-scoring matrix to trigger automated containment playbooks against surging digital payment anomalies.